

```
1.  #include <stdio.h>
2.
3.  int main()
4.  {
5.      int num;
6.      int marks;
7.      int year;
8.      int age;
9.
10.     printf("Enter a number: ");
11.     scanf("%d",&num);
12.
13.     if(num % 2 == 1)
        // ERROR: This condition checks for odd numbers.
        // CORRECTION: if(num % 2 == 0)
14.     printf("Number is Even\n");
        // ERROR: Even and odd messages are reversed.
15.     else
16.     printf("Number is Odd\n");
17.
18.     printf("\n");
19.
20.     printf("Enter marks: ");
21.     scanf("%d",&marks);
22.
23.     if(marks >= 90)
24.         printf("Grade B\n");
        // ERROR: Marks >= 90 should be Grade A.
        // CORRECTION: printf("Grade A\n");
25.     else if(marks >= 80)
26.         printf("Grade A\n");
        // ERROR: Marks 80–89 should be Grade B.
        // CORRECTION: printf("Grade B\n");
27.     else if(marks >= 70)
28.         printf("Grade C\n");
29.     else if(marks >= 60)
30.         printf("Grade D\n");
31.     else
32.         printf("Fail\n");
33.
34.     printf("\n");
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35.
36.     printf("Enter year: ");
37.     scanf("%d",&year);
38.
39.     if(year % 4 == 0)
        // ERROR: Leap year logic is incomplete.
        // CORRECTION:
        // if((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
40.         printf("Leap Year\n");
41.     else
42.         printf("Not Leap Year\n");
43.
44.     printf("\n");
45.
46.     printf("Enter age: ");
47.     scanf("%d",&age);
48.
49.     if(age > 18)
        // ERROR: People aged 18 and above can vote.
        // CORRECTION: if(age >= 18)
50.         printf("Not Eligible for Voting\n");
        // CORRECTION: printf("Eligible for Voting\n");
51.     else
52.         printf("Eligible for Voting\n");
        // CORRECTION: printf("Not Eligible for Voting\n");
53.
54.     printf("\n");
55.
56.     int a = 10;
57.     int b = 20;
58.
59.     if(a > b)
60.         printf("Smaller Number = %d\n", a);
        // ERROR: If a > b, then b is smaller.
        // CORRECTION: printf("Smaller Number = %d\n", b);
61.     else
62.         printf("Smaller Number = %d\n", b);
        // ERROR: If a <= b, then a is smaller.
        // CORRECTION: printf("Smaller Number = %d\n", a);
63.
64.     printf("\n");

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65.
66.   int x = 10;
67.   int y = 20;
68.   int z = 30;
69.
70.   int largest = x;
71.
72.   if(y < largest)
    // ERROR: Wrong comparison operator.
    // CORRECTION: if(y > largest)
73.   largest = y;
74.
75.   if(z < largest)
    // ERROR: Wrong comparison operator.
    // CORRECTION: if(z > largest)
76.   largest = z;
77.
78.   printf("Largest Number = %d\n", largest);
79.
80.   printf("\n");
81.
82.   int totalMarks = 250;
83.   int subjects = 3;
84.
85.   float average;
86.
87.   average = totalMarks / subjects;
    // ERROR: Integer division loses decimal precision.
    // CORRECTION: average = (float)totalMarks / subjects;
88.
89.   printf("Average = %.2f\n", average);
90.
91.   printf("\n");
92.
93.   int radius = 7;
94.
95.   float area;
96.
97.   area = 2 * 3.14 * radius * radius;
    // ERROR: Formula for area of circle is incorrect.
    // CORRECTION: area = 3.14f * radius * radius;
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98.
99.     printf("Circle Area = %.2f\n", area);
100.
101.     printf("\n");
102.
103.     int salary = 50000;
104.
105.     float bonus;
106.
107.     if(salary > 30000)
108.         bonus = salary * 0.02;
            // ERROR: Higher salary bonus may be too low.
            // CORRECTION (example): bonus = salary * 0.20;
109.     else
110.         bonus = salary * 0.20;
            // ERROR: Lower salary bonus may be too high.
            // CORRECTION (example): bonus = salary * 0.02;
111.
112.     printf("Bonus = %.2f\n", bonus);
113.
114.     printf("\n");
115.
116.     int month = 4;
117.
118.     switch(month)
119.     {
120.         case 1:
121.             printf("February\n");
                // ERROR: Month 1 is January.
                // CORRECTION: printf("January\n");
122.             break;
123.
124.         case 2:
125.             printf("March\n");
                // ERROR: Month 2 is February.
                // CORRECTION: printf("February\n");
126.             break;
127.
128.         case 3:
129.             printf("April\n");
                // ERROR: Month 3 is March.
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        // CORRECTION: printf("March\n");
130.     break;
131.
132.     case 4:
133.         printf("May\n");
        // ERROR: Month 4 is April.
        // CORRECTION: printf("April\n");
134.     break;
135.
136.     default:
137.         printf("Invalid Month\n");
138. }
139.
140. printf("\n");
141.
142. int n = 13;
143. int isPrime = 1;
144.
145. for(int i=2;i<n;i++)
146. {
147.     if(n % i == 0)
148.     {
149.         isPrime = 1;
        // ERROR: Number is not prime if divisible.
        // CORRECTION: isPrime = 0;
150.         break;
151.     }
152. }
153.
154. if(isPrime)
155.     printf("Not Prime\n");
        // ERROR: Prime logic is reversed.
        // CORRECTION: printf("Prime\n");
156. else
157.     printf("Prime\n");
        // CORRECTION: printf("Not Prime\n");
158.
159. printf("\n");
160.
161. int purchaseAmount = 5000;
162.

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163. float discount = 0;
164.
165. if(purchaseAmount > 1000)
166.     discount = 5;
        // ERROR: Discount should be percentage-based.
        // CORRECTION: discount = purchaseAmount * 0.05f;
167.
168. float finalAmount;
169.
170. finalAmount = purchaseAmount + discount;
        // ERROR: Discount should be subtracted.
        // CORRECTION: finalAmount = purchaseAmount - discount;
171.
172. printf("Final Amount = %.2f\n", finalAmount);
173.
174. printf("\n");
175.
176. int sum = 0;
177.
178. for(int i=1;i<=10;i++)
179. {
180.     sum = i;
        // ERROR: Overwrites previous value.
        // CORRECTION: sum += i;
181. }
182.
183. printf("Sum = %d\n", sum);
184.
185. printf("\n");
186.
187. int temperature = 35;
188.
189. if(temperature < 40)
        // ERROR: 35°C is generally considered hot.
        // CORRECTION: if(temperature < 20)
190.     printf("Cold Weather\n");
191. else
192.     printf("Hot Weather\n");
193.
194. printf("\n");
195.
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```
196. char grade = 'A';
197.
198. if(grade == 'A')
199.     printf("Average Performance\n");
    // ERROR: Grade A should indicate excellent performance.
    // CORRECTION: printf("Excellent Performance\n");
200. else
201.     printf("Excellent Performance\n");
    // ERROR: Non-A grades should not indicate excellent performance.
    // CORRECTION: printf("Average Performance\n");
202.
203. printf("\n");
204.
205. int length = 10;
206. int width = 5;
207.
208. int rectangleArea;
209.
210. rectangleArea = length + width;
    // ERROR: Rectangle area formula is incorrect.
    // CORRECTION: rectangleArea = length * width;
211.
212. printf("Rectangle Area = %d\n", rectangleArea);
213.
214. printf("\n");
215.
216. int percentage = 75;
217.
218. if(percentage >= 60)
219.     printf("Fail\n");
    // ERROR: Percentage >= 60 should pass.
    // CORRECTION: printf("Pass\n");
220. else
221.     printf("Pass\n");
    // ERROR: Percentage < 60 should fail.
    // CORRECTION: printf("Fail\n");
222.
223. printf("\n");
224.
225. printf("Program Completed Successfully\n");
226.
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227. return 0;  
228. }
```